

GANDHIMATHI (MATHI) PADMANABAN

University of Michigan-Dearborn

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RESEARCH PROFILE

I build and evaluate adaptable AI systems that are driven and constrained by physical measurements from multimodal data in safety-critical settings involving human interaction. My research sits at the intersection of computer vision, machine learning, and human-machine interaction, applied to transportation safety. My long-term research agenda is the reliability of those systems in use, for safer interaction between people and automation.

EDUCATION

- Aug 2026 **Ph.D., Industrial and Systems Engineering** - *University of Michigan-Dearborn, USA*
Dissertation: “Adaptable Machine Learning and Computer Vision Frameworks for Road Safety: Applications to Driver Behavior and Driver-Bicyclist Interaction Research” (*defended June 4, 2026*)
Advisor: [Dr. Fred Feng](#)
- 2021 **M.S., Human Centered Design and Engineering** - *University of Michigan-Dearborn, USA*
Thesis: “[Computational Human Performance Modeling using Queuing Network in an Open-Source Platform](#)”
Advisor: [Dr. Fred Feng](#)
- 2013 **B.E., Computer Science and Engineering** - *Anna University, India*
Thesis: “Automated Detection of Modifications in Software Requirement Traceability Links”
Advisor: Prof. Ramachandran Alagarsamy
- Certifications:** Rackham DEI Certificate (2025); PG Diploma in Computer Applications (2011)

RESEARCH EXPERIENCE

- 2022–2026 **Graduate Student Research Assistant** - *University of Michigan-Dearborn, USA*
Advisor/PI: [Dr. Fred Feng](#)
- Built and validated an automated measurement system that reports how closely vehicles pass cyclists from a single uncalibrated camera, using features derived from projective geometry (0.126 m MAE; re-fit on the Waymo Open Dataset, $R^2=0.812$).
 - Tested the physical derivation rather than assuming it: predicted what the fitted coefficient should equal under the pinhole model and confirmed the data recovers it (implied focal length 0.996 of the true value; segment level 0.974, CI [0.953, 1.004]).
 - Fine-tuned a Vision Transformer classifier that holds 89% accuracy at sites it has never seen with no retraining and **declines to answer under low confidence rather than misclassify**, on a dataset I curated, annotated, documented and prepared for public release.
 - Modelled driver behaviour from vehicle kinematics, and tested whether a **standardized certification test procedure represents** the driving behaviour it is used to certify.
 - Mentored undergraduate and master’s students in machine learning and Python data pipelines, including contributions to open-source tooling.
- 2011–13 **Research Assistant** - *Anna University, India*
Advisor/PI: Prof. Ramachandran Alagarsamy
- Contributed to software testing research group; led development of university applications including exam cell software, attendance management software and sports management application using full-stack development approaches.
 - Developed automated detection system for modifications in software requirement traceability links (Bachelor’s thesis)

PUBLICATIONS

Peer-Reviewed Works

1. **Gandhimathi Padmanaban**, Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. “A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles”. *WCX SAE World Congress Experience*. 2025. DOI: <http://dx.doi.org/10.4271/2025-01-8605>.
2. Rayane Moustafa, **Gandhimathi Padmanaban**, and Fred Feng. “Quantifying Drivers-Overtaking-Bicyclists with Surrogate Safety Measures Derived from High-Resolution Digital Lidar”. *TRB Annual Meeting*. 2026. URL: <https://par.nsf.gov/biblio/10679666>.

3. **Gandhimathi Padmanaban**, Nathaniel P. Jachim, Hala Shandi, Lilit Avetisyan, Garrett Smith, Howrah Hamoud, and Feng Zhou. “An Autonomous Driving System - Dedicated Vehicle for People with ASD and their Caregivers”. *AutomotiveUI '21 Adjunct: 13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. Association for Computing Machinery, 2021, pp. 142–147. DOI: <https://doi.org/10.1145/3473682.3480282>.

Papers Under Review & Preprints

1. **Gandhimathi Padmanaban**, Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. “A Machine Learning Framework to Identify Aggressive Driving Based on Vehicle Kinematics and Driver Pedal Operations”. *Submitted – Engineering Applications of Artificial Intelligence*.
2. **Gandhimathi Padmanaban**, Rayane Moustafa, and Fred Feng. “A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle”. *Submitted – IEEE Transactions on Intelligent Transportation Systems*. URL: <https://par.nsf.gov/biblio/10679667>.
3. **Gandhimathi Padmanaban** and Fred Feng. “Vision-Based Lateral Passing Distance Estimation from Bicycle-Mounted Cameras: A Projective Geometry Approach”. *In Preparation – IEEE Transactions on Image Processing*.
4. **Gandhimathi Padmanaban** and Fred Feng. “A Multi-View Vehicle Image Dataset and Two-Stage Pipeline for Fine-Grained Vehicle-Type Recognition at Ground Level”. *In Preparation for CVPR*. 2027.

TECHNICAL EXPERTISE

<i>AI/ML Methods</i>	Deep learning (CNN, RNN, Transformers, ViT); supervised learning and ensembles; sequential and dynamical system modelling; Bayesian methods; out-of-distribution detection and cross-site generalization; conformal and selective prediction (in progress); geometry-informed computer vision
<i>Perception & Sensing</i>	Multi-object detection and tracking (RT-DETR, ByteTrack, YOLOv5); multimodal camera and LiDAR pipelines; sensor fusion; vehicle kinematics modelling; ROS; Vision-Language Models
<i>Evaluation, Data & Standards</i>	Evaluation and measurement validity; benchmark and test-procedure critique; dataset construction, annotation, documentation and public release; cross-platform evaluation
<i>Human Systems & HCAI</i>	Human performance modelling (queueing networks); human-in-the-loop handoff design; abstention and selective prediction so a system defers rather than guess; human-automation interaction on the machine side; decision-support evaluation
<i>Software & Platforms</i>	Primary: Python (PyTorch, TensorFlow, scikit-learn, OpenCV, pandas), Git, Docker, L ^A T _E X; Working: Julia, R, MATLAB, C#, SQL, UI frameworks, full-stack web; HPC: Great Lakes cluster (Slurm); production software delivery in industry
<i>Foundations</i>	Optimization; statistical learning; probability and queueing theory; multivariable calculus; mathematical modelling; projective geometry; dynamical systems
<i>Research Practice</i>	Experimental design; statistical analysis; scientific writing; peer review; IRB protocols; cross-validation; research project management and stakeholder communication
<i>Domains</i>	Human-machine systems; safety-critical and decision-support AI; autonomous and connected vehicles; transportation safety; naturalistic driving studies; vulnerable road user safety; human factors

AWARDS AND GRANTS

- **Student Visionary Award** - International Forum on Research Excellence Conference (IFoRE' 25), Sigma Xi-The Scientific Research Honor Society, 2025
- **Upsilon Pi Epsilon (UPE) Scholarship** - Awarded for exceptional academic performance, extracurricular involvement, and leadership within the computing community, 2024
- **Global Finalist - NASA Space Apps Challenge**, 2023
- **Irma M. Wyman Scholar**, *Center for the Education of Women (CEW+)*, *University of Michigan*, 2020-2021 (\$11,500)
- Non-Resident Graduate Student Scholar, *University of Michigan-Dearborn*, 2020-2021 (\$13,000)
- Deloitte Hackathon Special Mention, 2017 | Syncfusion Hackathon 2nd Place, 2015 (INR 35,000)

TALKS AND PRESENTATIONS

Talks

- **WocCode – University of Michigan Ann Arbor**, Feb 2024
- **NASA SpaceApps Experience: DigitwiML – Open-Source Project to model Digital Twin of C.elegans in Space**
- **University of Michigan–Dearborn**, Sep 2021
- Guest Talk: IMSE 501 Human Factors & Ergonomics

Conference Presentations

- **Transportation Research Board Annual Meeting**, Jan 2026 (*Poster*)
A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle
- **Transportation Research Board Annual Meeting**, Jan 2026 (*Presentation*)
Quantifying Drivers-Overtaking-Bicyclists with Surrogate Safety Measures Derived from High-Resolution Digital Lidar (*Moustafa, Padmanaban, Feng*)
- **International Forum on Research Excellence (IFoRE' 25)**, Oct 2025 (*Workshop Session*)
Hybrid Physics-Data Modeling for Sustainable Transportation: Bridging Classical Models and Modern AI
- **WCX SAE World Congress Experience**, Apr 2025 (*Paper Presentation*)
A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles
- **Automotive User Interfaces (AutoUI) Conference**, Sep 2021 (*Paper Presentation*)
An Autonomous Driving System – Dedicated Vehicle for People with ASD and their Caregivers

RESEARCH SOFTWARE & OPEN SOURCE PROJECTS

- **DigitwiML** (NASA SpaceApps Global Finalist, 2023): Machine learning platform for digital twin modeling of biological dynamics in space environments. First application of data-driven dynamical modeling beyond transportation, demonstrating interest in scientific ML for physical systems - [GitHub](#)
- **Geometry-informed overtaking tracker** - Automated measurement of vehicle-bicyclist interactions with a physical plausibility check on detector output (open-source, Python) - [GitHub](#)
- **Vehicle Type Classifier** - Injury-risk vehicle classification that abstains under low confidence rather than misclassify (open-source, Python) - [GitHub](#)

PROFESSIONAL EXPERIENCE (INDUSTRY)

2017–18	Consultant / Lead Front End Developer , Deloitte (Offices of the US) – Bangalore, India Led the technical UI team and owned stakeholder requirements through delivery.
2015–17	Programmer Analyst , Cognizant – Chennai, India Built production learning-platform software for Pearson Education.
2013–15	Software Engineer Level-II , Syncfusion – Chennai, India

TEACHING AND MENTORING

2026	Teaching Assistant - <i>Python as an Engineering Tool</i> , University of Michigan-Dearborn Intensive Python module for visiting engineering students in the ESIAE exchange program.
2024–Now	Certified Instructor - <i>The Carpentries</i> Hands-on computational methods workshops (Python data analysis, Git version control, Unix shell, OpenRefine) for researchers in biosciences, library science, and data-intensive fields. University of Tennessee Knoxville (May 2025) CZI Foundation (Nov 2024) University of Michigan (Mar 2024)
2023–Now	Instructor & Mentor - <i>WoCCode</i> - University of Michigan-Ann Arbor Machine Learning and Python workshop: materials open-sourced. Machine Learning Workshop (Feb 2024) Summer 2023 Boot Camp
2010–12	Student Instructor / Teaching Assistant - Anna University, India Courses: Artificial Intelligence, Probability and Queuing Theory, Transforms and PDEs, Systems Software Laboratory

SERVICE AND LEADERSHIP

Leadership: President, [Upsilon Pi Epsilon](#) Michigan Beta Chapter (2024-2025) • Member: [SAE International](#) (AI in Simulation Task Force), [Sigma Xi](#), [Alpha Pi Mu](#), [Society of Industrial and Applied Mathematics](#), [ACM](#), [HFES](#), [INFORMS](#) | Selected, [INFORMS WORMS To My Younger Self](#) mentee cohort (2026)

Reviewer: [IEEE T-ITS](#), [TRBAM 2026](#), [ICIS 2025](#), [AutoUI 2024](#), [CHI 2024](#), [AMCIS 2024](#), [CUI 2024](#), [IMX 2024](#), [DIS 2024](#)

Community: [MIDAS AI Summit](#) Student Volunteer (2023,2025) • [WoC|CodeNova Hackathon](#) Judge (2025) • [Woc-Code Mentor](#) • [Carpentries Certified Instructor & Lesson Maintainer](#) • [NASA SpaceApps](#) & [MHacks](#) Judge • [STEM High School Tutor](#) (2010-13)